



Biological Bases of Behaviour

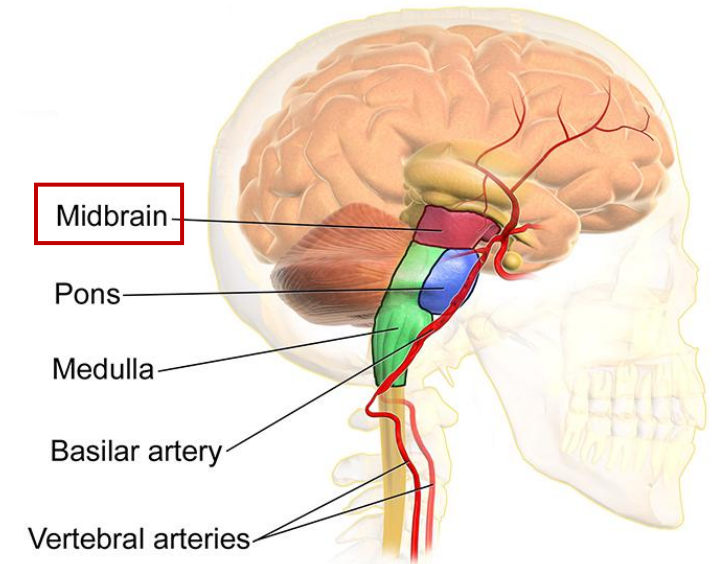
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RECAP

- Cortex
 - Frontal lobe: Broca Aphasia (difficulty in using expressive language)
 - Temporal lobe: Wernicke's aphasia (word salad/ difficult in word comprehension)
- Basal Ganglia (decision making and coordination)
 - Addiction
 - Alzheimer's
 - Parkinson's
 - Huntington's (Decay of brain nerve cells over time)
 - Tourette syndrome
- Limbic System
 - Thalamus (relay station for sensory except smell): Dyslexia
 - Hypothalamus (controls emotions): Hormones issues/ Depression
 - Cingulate cortex: Dyslexia (selective attention)
 - Hippocampus: Alzheimer (memory)
 - Amygdala: Klüver-Bucy syndrome

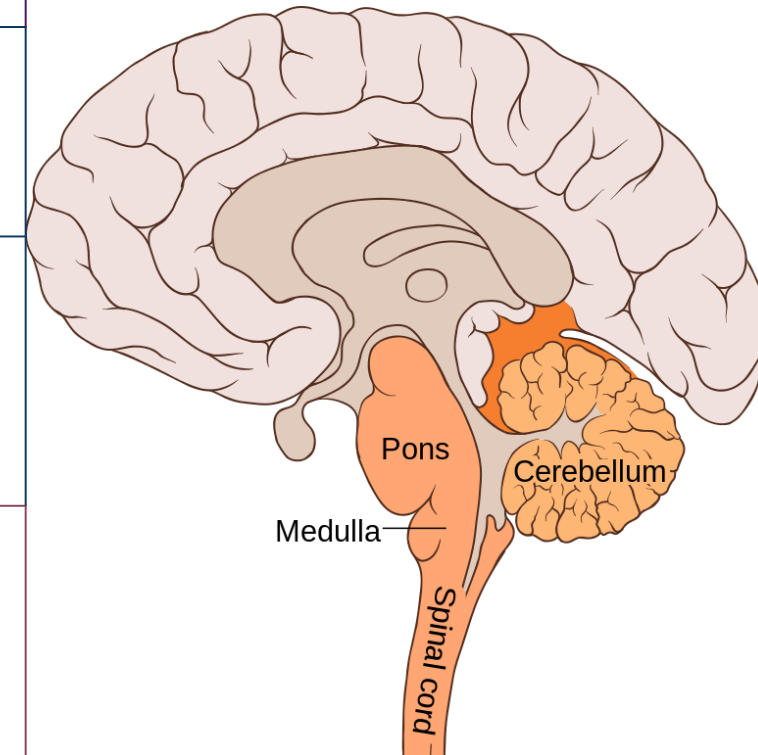
Structure of the Brain: Midbrain

- Situated below Thalamus
- Processes visual and auditory signals
- Motor movement and reflexes
- Regulates alertness, wakefulness, sleep-wake cycles
- Pain signals
- Damage: Parkinson's disease, issues with sleep



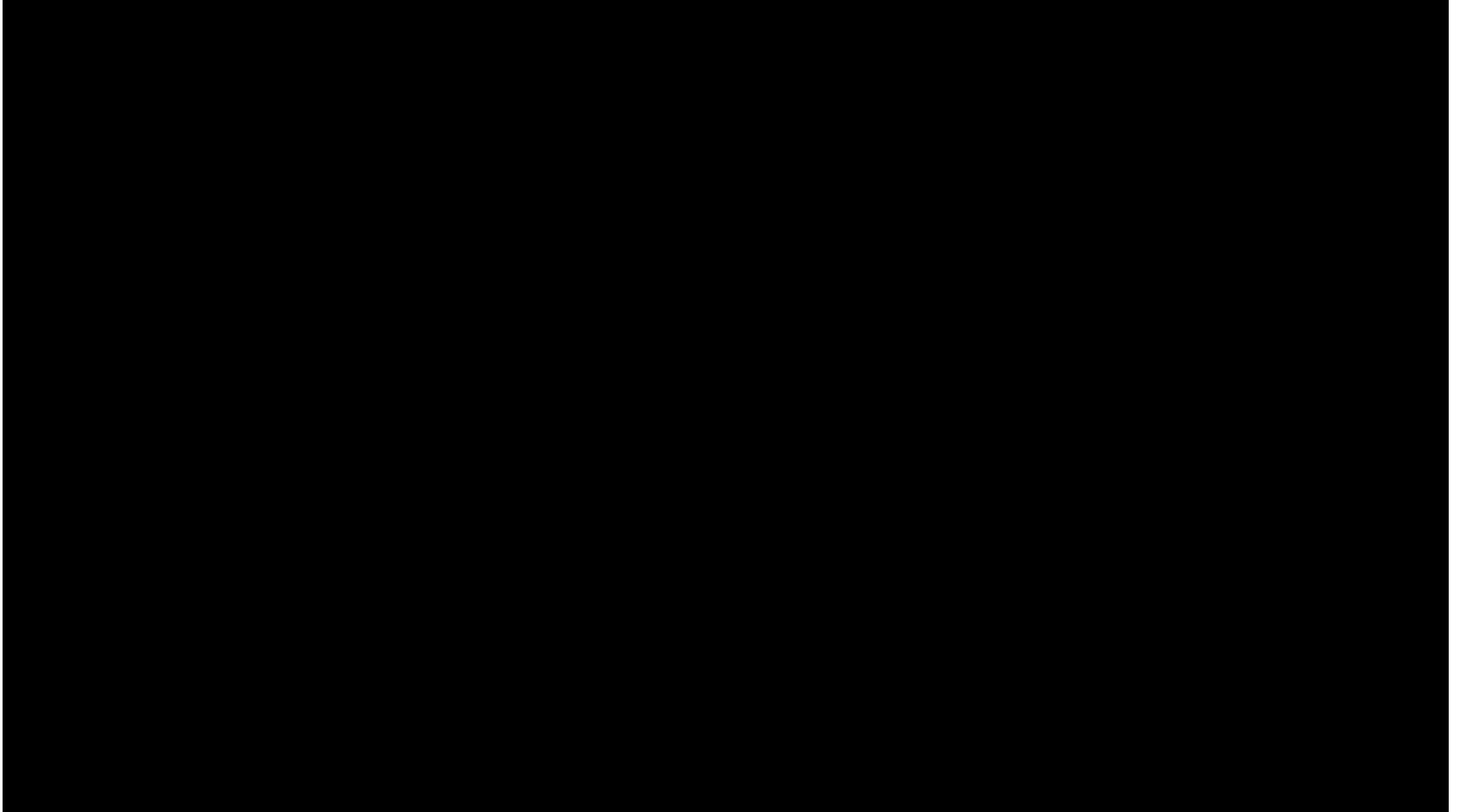
Structure of the Brain: Hindbrain

Hindbrain Structure	Location & Role	Consequences of Malfunction	Mnemonic
Medulla	- Top of spinal column - Heartbeat, breathing, swallowing - Sensory nerves crossover	Damage can be fatal; disrupts vital functions like heartbeat, breathing, swallowing	Medulla = Must survive (life functions)
Pons	- Above medulla - Sleep, dreaming, arousal - Motor nerve crossover (left-right body movements)	Impaired coordination of movements	Pons = Point to coordination (bridge, motor)
Cerebellum	- Behind pons - Controls (in)voluntary, rapid fine motor movement, balance, posture, learned motor skills	- Tremors, unsteady walk, slurred speech, dizziness, muscle weakness (e.g., spinocerebellar degeneration) - Symptoms similar to alcoholic intoxication - Schizophrenia, autism spectrum disorder	Cerebellum = Coordination centre



 Cerebellum aka little brain

NEUROPLASTICITY: Brains ability to reorganise its structures, function, and neural connections



Neuroplasticity

Brains ability to change its structure and function in response to experience , learning, environment, and injury

Why it Matters?

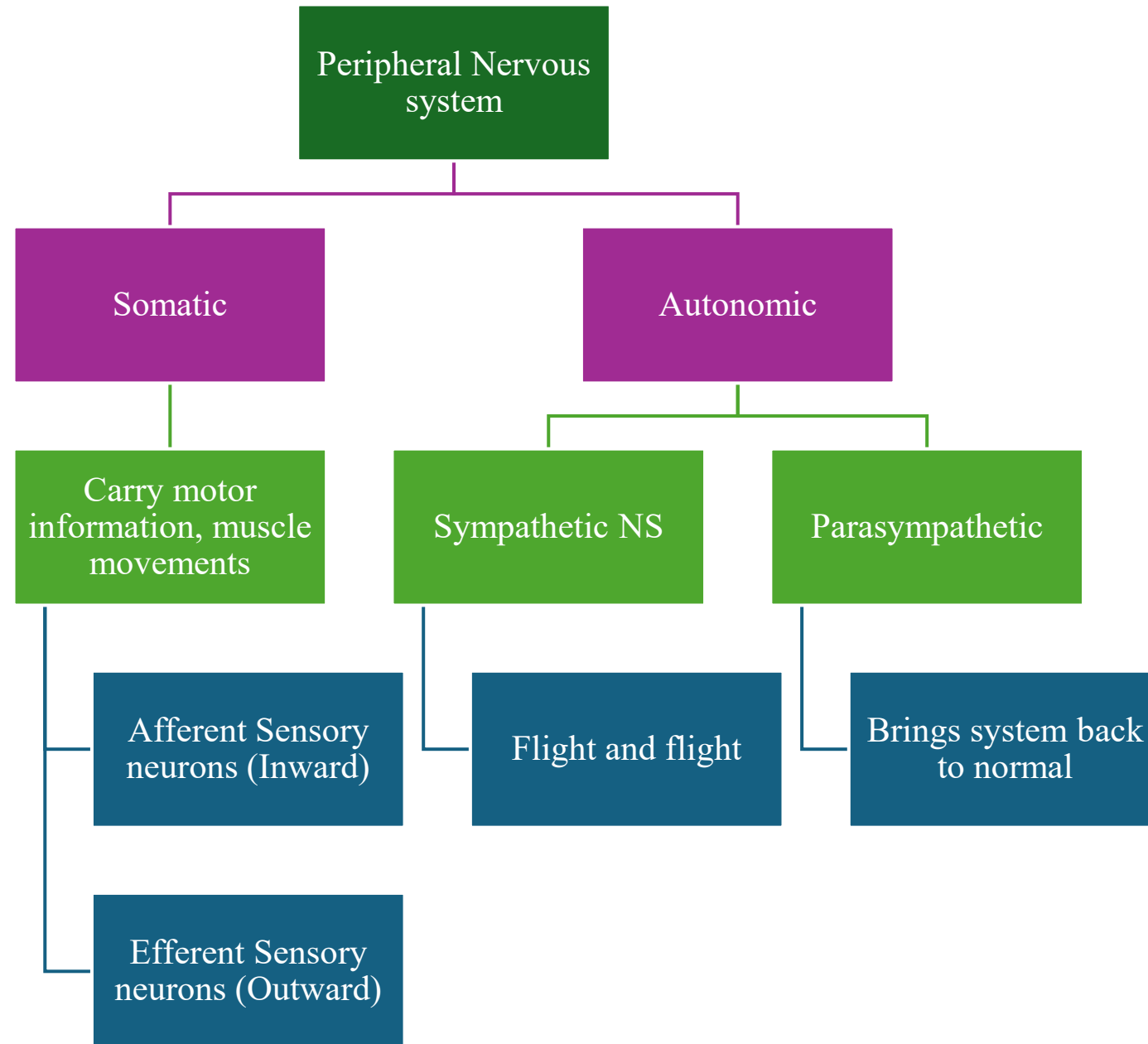
- Basic learning and memory
- Recovery from injury
- Mechanism behind rehabilitation and behaviour change

Types of neuroplasticity

- Experience dependent plasticity (practice and exposure)
- Developmental plasticity: heightened plasticity during critical periods in childhood
- Injury-induced plasticity: Reorganisation following brain injury

Clinical and applied relevance

- Rehabilitation: Stroke, traumatic brain injury, recovery rely heavily on neuroplastic mechanisms
- Education and skill learning: Teaching strategies leverage plasticity through spaced practice and multimodal learning



The Autonomic Nervous System

Sympathetic division

- Most active when you are angry, afraid, or aroused
- **Fight-or-flight response**
- Increases heart rate and breathing
- Stops digestion

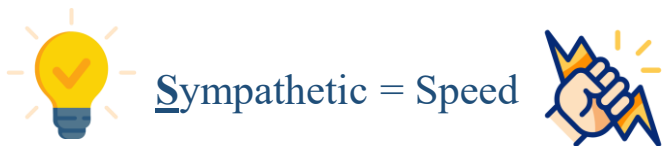
Sympathetic
division (arousing)

Pupils dilate	EYES	Pupils contract
Decreases	SALIVATION	Increases
Perspires	SKIN	Dries
Increases	RESPIRATION	Decreases
Accelerates	HEART	Slows
Inhibits	DIGESTION	Activates
Secrete stress hormones	ADRENAL GLANDS	Decrease secretion of stress hormones

Parasympathetic
division (calming)

Parasympathetic division

- **Calms body**
- Produces effects opposite to those of the sympathetic division
- Reduces heart rate and breathing
- Restores digestion



Sympathetic = Speed

The Somatic Nervous System

- Somatic Nervous System: voluntary nervous system
- Consists of neurons that communicate between the body and the brain
- **Afferent (sensory) neurons [inward]**
 - Neurons that carry messages from sense organs to spinal cord
- **Efferent (motor) neurons [outward]**
 - Neurons that carry messages from the spinal cord or brain to muscles and glands
- Also controls reflex which is an involuntary muscle response
 - **Interneurons** connect the afferent neurons to the efferent neurons



Afferent neurons access the spinal cord
Efferent neurons exit

- Disease affecting motor neurons are motor neuron disease: progressive muscle weakness and waste



Knee-jerk¹⁰ reflex

Key Stuff to Lock In



- Brain
- Consciousness
- Nervous system
- Central Nervous system
- Brain
- Spinal cord
- Cranium or skull
- Cerebrum
- Cerebellum
- Diencephalon
- Midbrain
- Pons
- Medulla Oblongata

- Frontal
- Parietal
- Temporal
- Occipital
- Thalamus
- Hypothalamus
- Hippocampus
- Amygdala
- Afferent spinal nerves
- Efferent spinal nerves
- Peripheral Nervous system
- Somatic
- Autonomic

- Sympathetic
- Parasympathetic
- Broca Aphasia
- Wernicke's aphasia
- Addiction
- Alzheimer's
- Parkinson's
- Huntington's
- Tourette syndrome
- Dyslexia
- Cingulate cortex
- Klüver-Bucy syndrome
- Neuroplasticity

References

- Ciccarelli, S. K., White, J. N., & Misra, G. (2017). *Psychology* (5th ed.). Pearson Education India.
- Richards, L. J., Kilpatrick, T. J., & Bartlett, P. F. (1992). De novo generation of neuronal cells from the adult mouse brain. *Proceedings of the National Academy of Sciences*, 89(18), 8591–8595. <https://doi.org/10.1073/pnas.89.18.8591>

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